Homework 5

Problem 1

In a certain region, times (minutes) between occurrences of earthquakes (of any magnitude) have a distribution with percentiles displayed in the table below.

1. Construct a spinner corresponding to this distribution.
2. What percent of times are between 26.8 and 110.0 minutes?
3. Let F be the cdf. Evaluate and interpret $F(42.8)$.
4. Let Q be the quantile function. Evaluate and interpret $Q(0.9)$.
5. Sketch (by hand) a histogram of this distribution.

Problem 2

Recall Example 2.6. Regina and Cady are meeting for lunch. Suppose they each arrive uniformly at random at a time between noon and 1:00, independently of each other. Record their arrival times as minutes after noon, so noon corresponds to 0 and 1:00 to 60. Recall the sample space from Example 2.2 and assume a uniform probability measure P.

Let R be Regina's arrival time and Y Cady's. Then $W = |R - Y|$ is the amount of time (minutes) the first person to arrive waits for the second person to arrive. It can be shown that W has pdf

$$f_W(w) = (2/3600)(60 - w), \qquad 0 < w < 60.$$

Percentile	Value (minutes)
10th	12.6
20th	26.8
30th	42.8
40th	61.3
50th	83.2
60th	110.0
70th	144.5
80th	193.1
90th	276.3

1. Let F be the cdf. Evaluate and interpret $F(15)$.
2. Find an expression for the cdf of W. Set up an integral, but sketch a picture and use geometry to compute.
3. Find and interpret the 25th percentile of W. (You can do the next part first if you want, but it might help to start with a specific number like in this part.)
4. Find the quantile function of W.
5. Sketch a spinner corresponding to the distribution of W. Label the 25th, 50th, and 75th percentiles.

Problem 3

Continuing Problem 2. For this problem you should set up integrals when appropriate but you can use software to compute them.

1. Explain in full detail how you could use simulation to approximate
   a. $\mathrm{E}(W)$
   b. $\mathrm{E}(W^2)$
   c. $\mathrm{Var}(W)$ (explain how you can do it directly without using the previous part.)
2. Compute and interpret $\mathrm{E}(W)$.
3. Compute $\mathrm{E}(W^2)$.
4. Compute $\mathrm{Var}(W)$
5. Compute and interpret $\mathrm{SD}(W)$.
6. Compute the probability that W is more than 1 SD away from its mean.

Problem 4

Lifetimes of a certain type of component are Exponentially distributed with mean 20 thousand hours.

1. Approximate the probability that the lifetime, rounded to the nearest thousand hours, of a randomly selected component is 30 thousand hours.
2. Compute the probability that a randomly selected component has a lifetime greater than 30 thousand hours.
3. Compute the standard deviation of component lifetimes.
4. Suppose a device contains 2 such components connected in series, so that the system functions only if both of the components are functioning. Suppose the component lifetimes are independent of each other. Compute the probability that the *device* has a lifetime greater than 30 thousand hours.
5. Let T be the lifetime of the *device*. Compute $\mathrm{P}(T > t)$.
6. Identify the distribution of T by name, including the values of any relevant parameters.